

A closed form for the Apéry $\zeta(2)$ companion, and a reduction for its minimal form

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Abstract

Let $S(n, k) = \binom{n}{k}^2 \binom{n+k}{k}$ and let (B_n) be the second solution of Apéry's $\zeta(2)$ recurrence. We prove $5B_n = \sum_k S(n, k) \left(2 \sum_{m \leq n} \frac{(-1)^{m-1}}{m^2} + \sum_{m \leq k} \frac{(-1)^{n+m-1}}{m^2 \binom{n}{m} \binom{n+m}{m}} \right)$ by an explicit certificate chain whose last step degenerates, remarkably, to the identity $W(n, k) = 3 - \frac{3(n+1)}{n+1-k}$ together with the binomial theorem. The depth-one harmonic (“minimal”) form of the companion is shown to be equivalent to a single inhomogeneous telescoping identity, which we leave open with a precise record of the certificate classes that provably cannot settle it.

1 Statement

Let

$$S(n, k) = \binom{n}{k}^2 \binom{n+k}{k}, \quad a_n = \sum_{k=0}^n S(n, k),$$

$$(\mathbf{L} u)_n = (n+1)^2 u_{n+1} - (11n^2 + 11n + 3)u_n - n^2 u_{n-1},$$

so $\mathbf{L} a = 0$, and let (B_n) be the second solution: $\mathbf{L} B = 0$, $B_0 = 0$, $B_1 = 1$; classically $B_n/a_n \rightarrow \zeta(2)/5$. Put

$$g_m(n) = \frac{1}{m^2 \binom{n}{m} \binom{n+m}{m}} = \frac{((m-1)!)^2 (n-m)!}{(n+m)!}, \quad \tau(n, k) = \sum_{m=1}^{\min(k, n)} (-1)^{n+m-1} g_m(n),$$

$$e(n) = \sum_{m=1}^n \frac{(-1)^{m-1}}{m^2}, \quad w(n, k) = 2e(n) + \tau(n, k).$$

Theorem 1. *For all $n \geq 0$, $\sum_{k=0}^n S(n, k) w(n, k) = 5B_n$.*

This is the $\zeta(2)$ analogue of the classical representation of the $\zeta(3)$ companion (van der Poorten §8); we have not located it in the literature in this form. The alternating factor $(-1)^n$ inside the inner sum is essential.

2 The miracle, and the two shifts of the tail

Write $\sigma = (-1)^n$ and $\gamma(n, k) = (-1)^k (k!)^2 (n-k)! / (n+k+1)!$.

Lemma 1. For $0 \leq k \leq n$,

$$\tau(n+1, k) - \tau(n, k) = \frac{2\sigma}{n+1} \left(\gamma(n, k) - \frac{1}{n+1} \right),$$

and for $0 \leq k \leq n-1$, $\tau(n-1, k) - \tau(n, k) = \frac{2\sigma}{n} \left(\gamma(n-1, k) - \frac{1}{n} \right)$.

Proof. $(-1)^m [g_m(n+1) + g_m(n)] = 2(n+1) (-1)^m \frac{((m-1)!)^2 (n-m)!}{(n+m+1)!}$, while

$$\gamma(n, m) - \gamma(n, m-1) = (-1)^m \frac{((m-1)!)^2 (n-m)!}{(n+m+1)!} [m^2 + (n+1-m)(n+1+m)],$$

and $m^2 + (n+1-m)(n+1+m) = (n+1)^2$. Summing over $m \leq k$ telescopes; $\gamma(n, 0) = 1/(n+1)$. The second identity is the first at $n-1$. $\square$

Since $e(n \pm 1) - e(n)$ cancels the constant terms,

$$w(n+1, k) = w(n, k) + \frac{2\sigma}{n+1} \gamma(n, k), \quad w(n-1, k) = w(n, k) + \frac{2\sigma}{n} \gamma(n-1, k). \quad (1)$$

3 Proof of Theorem 1

Let $F(n, k) = S(n, k)w(n, k)$ and $\rho(n, k) = k^2 + k(1+6n) - 4 - 15n - 11n^2$, and set

$$G(n, k) = \frac{k^3 \rho(n, k)}{n(n+1)^2} \binom{n+1}{k}^2 \binom{n+k-1}{k}, \quad U(n, k) = (-1)^{n+k} \binom{n}{k}.$$

Step 1 (Zeilberger certificate). For all $k \geq 0$, $n \geq 1$: $(L S(\cdot, k))_n = G(n, k+1) - G(n, k)$. After the absorptions $\binom{n}{k}(n+1) = \binom{n+1}{k}(n+1-k)$ etc., this is a single polynomial identity in $\mathbb{Q}[n, k]$, verified by expansion; the shell form of G makes it valid at every cell, including $k = n, n+1$.

Step 2 (split). By (1) and the conversions $S(n+1, k)\gamma(n, k) = (-1)^k \binom{n+1}{k} / (n+1-k)$, $S(n-1, k)\gamma(n-1, k) = (-1)^k \binom{n-1}{k} / (n+k)$: for $0 \leq k \leq n-1$,

$$(L F(\cdot, k))_n = (L S(\cdot, k))_n w(n, k) + U(n, k) \left[\frac{2(n+1)^2}{(n+1-k)^2} - \frac{2(n-k)}{n+k} \right].$$

Step 3 (Abel). Summing $k = 0, \dots, n+1$, using Step 1, $G(n, 0) = 0$, $G(n, n+2) = 0$, $\Delta_k \tau = (-1)^{n+k} g_{k+1}$ and $S(n, k+1)g_{k+1}(n) = \binom{n}{k+1} / (k+1)^2$, the sum of Step 2 collapses to

$$(L v)_n = \sum_{k=0}^n (-1)^k \binom{n}{k} W(n, k) \cdot (-1)^n - 2(n+1)^2 + B_1(n) + B_2(n),$$

where $v_n = \sum_k F(n, k)$, $W = \frac{2(n+1)^2}{(n+1-k)^2} - \frac{2(n-k)}{n+k} + \frac{k\rho(n, k)}{(n+1-k)^2(n+k)}$, and B_1, B_2 are the two boundary cells $k = n, n+1$. Explicitly $B_1 = 2(n+1)^2$, and $B_2 = -1 + 2(n+1) \binom{2n+2}{n+1} (n!)^2 / (2n+1)! = -1 + 4 = 3$.

Step 4 (the collapse). As rational functions,

$$W(n, k) = 3 - \frac{3(n+1)}{n+1-k}$$

— both the double pole at $k = n+1$ and the pole at $k = -n$ cancel identically, the former by the boundary identity $\rho(n, n+1) = -2(n+1)(2n+1)$. Hence, by $\sum_k (-1)^k \binom{n}{k} = 0$ ($n \geq 1$) and

$$\sum_{k=0}^n \frac{(-1)^k \binom{n}{k}}{n+1-k} \stackrel{k \mapsto n-k}{=} (-1)^n \sum_{k=0}^n \frac{(-1)^k \binom{n}{k}}{k+1} = \frac{(-1)^n}{n+1},$$

(the last step by $\binom{n}{k}/(k+1) = \binom{n+1}{k+1}/(n+1)$ and the binomial theorem), we get $\sum_k (-1)^k \binom{n}{k} W = 3(-1)^{n+1}$ and therefore $(Lv)_n = -3 - 2(n+1)^2 + 2(n+1)^2 + 3 = 0$ for $n \geq 1$.

With $v_0 = 0$, $v_1 = 5 = 5B_1$ and $(n+1)^2 \neq 0$, induction gives $v_n = 5B_n$. $\square$

4 The minimal form: an exact reduction

Let $w_{\min} = \frac{1}{5}[H_n^{(2)} + H_k(2H_k - H_{n-k} - H_n)]$, the depth-one harmonic weight fitted in the program's ledger, with $\sum_k S w_{\min} = B_n$ verified for $n \leq 25$.

Proposition 1. $\sum_k S(n, k)w_{\min}(n, k) = B_n$ for all n if and only if

$$(LA)_n = -a_{n+1} - a_{n-1}, \quad A_n := \sum_{k=0}^n S(n, k) H_k(2H_k - H_{n-k} - H_n). \quad (2)$$

Proof. $L(H_n^{(2)}a)_n = a_{n+1} + a_{n-1}$ exactly (the shifts of $H^{(2)}$ produce $1/(n+1)^2$, $-1/n^2$ against the leading coefficients), so $L(\sum_k S w_{\min}) = \frac{1}{5}[(a_{n+1} + a_{n-1}) + (LA)_n]$; initial values $A_0 = 0$, $A_1 = 2$ match. $\square$

Identity (2) is true (verified exactly for $n \leq 25$) and open. Three certificate classes provably fail to establish it: the decoupled Gosper template (a sharp degree obstruction), one-shot coupled telescopes over the harmonic letter algebra on $\{k, n-k, n, n+k\}$ in two generous denominator/degree regimes, and the same augmented by a proved residue-null library at levels $n-1, n, n+1$. The obstruction has a structural face: the telescoping image of the summand's shell never leaves that shell, while the alternating shell $(-1)^{n+k} \binom{n}{k}$ — exactly what powers the proof of Theorem 1 — can enter only through a tail in the weight, and $w_{\min}$ has none.

Remark 1. All certificates and every step above were verified in exact rational arithmetic (all cells, $n \leq 25$ –30), and Steps 1 and 4 additionally as symbolic rational identities. Scripts: `work/z2cf/` in the program repository.